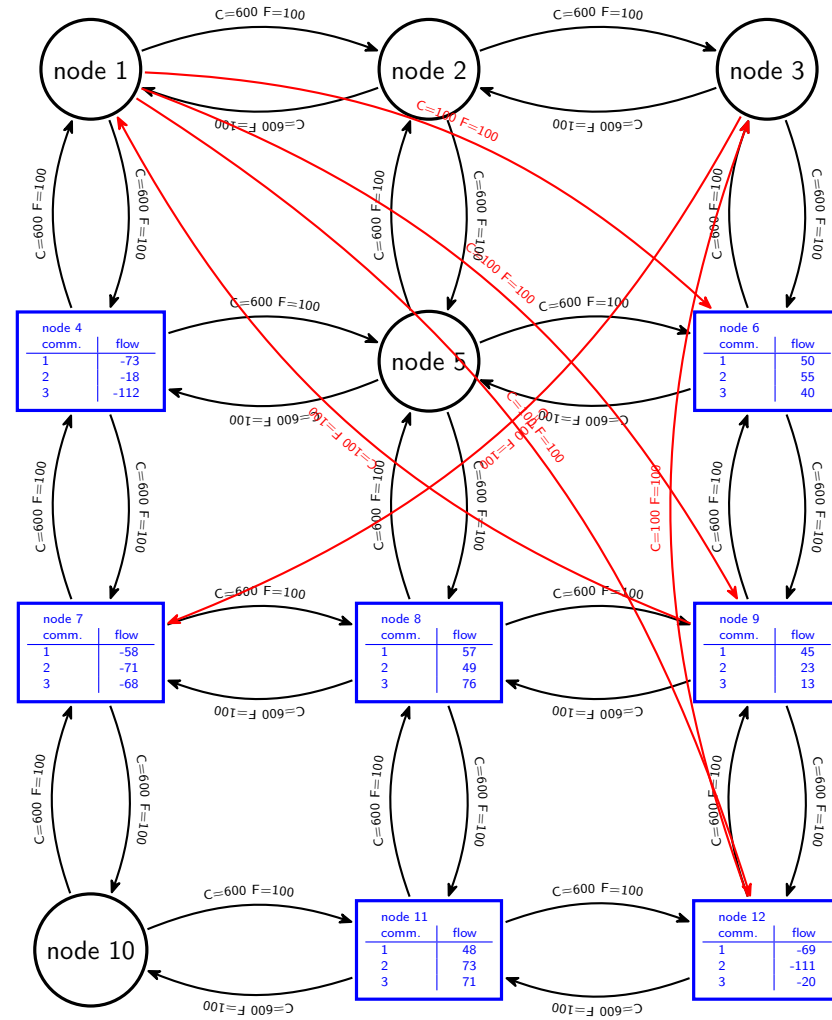


1 Generated network based on grid topology



2 Notes

- Black arcs are part of the base, grid topology.
- Red arcs are additional, randomly generated.
- $C = x$: overall capacity of arc is equal to x .
- $F = y$: fixed cost of arc is equal to y .
- See below for capacities and variable costs per arc, per commodity.
- Circular nodes are transit nodes.
- Rectangular nodes are source ($flow > 0$) or sink ($flow < 0$) nodes for stated commodities.

3 Capacities and variable costs per arc, per commodity

Arc	Comm.	Cap.	Var. cost	Comm.	Cap.	Var. cost	Comm.	Cap.	Var. cost
(1, 2)	1	200	10	2	200	10	3	200	10
(1, 4)	1	200	10	2	200	10	3	200	10
(1, 6)	1	100	10	2	100	10	3	100	10
(1, 9)	1	100	10	2	100	10	3	100	10
(1, 12)	1	100	10	2	100	10	3	100	10
(2, 1)	1	200	10	2	200	10	3	200	10
(2, 3)	1	200	10	2	200	10	3	200	10
(2, 5)	1	200	10	2	200	10	3	200	10
(3, 2)	1	200	10	2	200	10	3	200	10
(3, 6)	1	200	10	2	200	10	3	200	10
(3, 7)	1	100	10	2	100	10	3	100	10
(4, 1)	1	200	10	2	200	10	3	200	10
(4, 5)	1	200	10	2	200	10	3	200	10
(4, 7)	1	200	10	2	200	10	3	200	10
(5, 2)	1	200	10	2	200	10	3	200	10
(5, 4)	1	200	10	2	200	10	3	200	10
(5, 6)	1	200	10	2	200	10	3	200	10
(5, 8)	1	200	10	2	200	10	3	200	10
(6, 3)	1	200	10	2	200	10	3	200	10
(6, 5)	1	200	10	2	200	10	3	200	10
(6, 9)	1	200	10	2	200	10	3	200	10
(7, 4)	1	200	10	2	200	10	3	200	10
(7, 8)	1	200	10	2	200	10	3	200	10
(7, 10)	1	200	10	2	200	10	3	200	10
(8, 5)	1	200	10	2	200	10	3	200	10
(8, 7)	1	200	10	2	200	10	3	200	10
(8, 9)	1	200	10	2	200	10	3	200	10
(8, 11)	1	200	10	2	200	10	3	200	10
(9, 1)	1	100	10	2	100	10	3	100	10
(9, 6)	1	200	10	2	200	10	3	200	10
(9, 8)	1	200	10	2	200	10	3	200	10
(9, 12)	1	200	10	2	200	10	3	200	10
(10, 7)	1	200	10	2	200	10	3	200	10
(10, 11)	1	200	10	2	200	10	3	200	10
(11, 8)	1	200	10	2	200	10	3	200	10
(11, 10)	1	200	10	2	200	10	3	200	10
(11, 12)	1	200	10	2	200	10	3	200	10
(12, 3)	1	100	10	2	100	10	3	100	10
(12, 9)	1	200	10	2	200	10	3	200	10
(12, 11)	1	200	10	2	200	10	3	200	10